DNSC 6330: Responsible Machine Learning

Capstone Presentation — Student Rubric & Self-Assessment Guide

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How to Use This Document

This rubric shows you exactly what judges will look for when evaluating your capstone presentation. Read each question's performance descriptors before you rehearse. Use the self-check checklist at the bottom of each section to confirm your presentation is ready. The second half of this document includes presentation tips written specifically for your three judges and a suggested time allocation for your 15-minute slot.

The grading scale is straightforward: Exemplary earns full or near-full marks, Proficient earns partial credit, Developing earns minimal credit, and Inadequate earns zero. Judges score each question independently and award partial credit within bands at their discretion.

Points at a Glance

Capstone Question	Points Available
Q1 — Optimization Objective	**5**
Q2 — Known Failure Modes	**10**
Q3 — Subgroup Error Measurement	**10**
Q4 — Residual Risks After Mitigation	**3**
Q5 — Deployment Defensibility	**2**
TOTAL	**30**

Section-by-Section Rubric

Each table below shows you the four performance levels for that question, the criteria judges are evaluating, and a self-check list to use during rehearsal.

Q1 — Optimization Objective (5 points)

Judges want to know: what is the model optimizing for, who decided, and what are the trade-offs?

Criterion	Exemplary 4–5 pts	Proficient 3–4 pts	Developing 1–2 pts	Inadequate 0 pts
Metric & threshold	Named precisely with threshold (e.g., F1 ≥ 0.75 at 0.5 cutoff)	Named clearly; threshold mentioned but not fully justified	Named but with no threshold or justification	Not stated, or inappropriate metric chosen
Business rationale	Compelling, specific to HMDA lending; connects to applicant consequences	Evident and relevant; business context present but briefly stated	Generic (e.g., 'accuracy is standard'); lending context not engaged	No rationale provided

Trade-off analysis	Explicitly discussed with affected group named	Mentioned but not fully developed	Absent or only implied	No trade-off analysis of any kind

Before You Present — Self-Check

- My slide deck states the metric and threshold explicitly.
- I can explain why this metric fits a mortgage lending context.
- I can name at least one group that bears the cost of errors under this objective.
- I can articulate the precision-recall or fairness-accuracy trade-off in plain language.

Q2 — Known Failure Modes (10 points)

Judges want to know: where does your model break down, and who is harmed when it does?

Criterion	Exemplary 8–10 pts	Proficient 6–7 pts	Developing 3–5 pts	Inadequate 0–2 pts
Number of modes	At least two distinct, well-evidenced modes	Two modes identified with reasonable evidence	One or two modes named in general terms	No specific failure modes identified
Evidence quality	Drawn directly from team's own analysis	Mostly from own analysis; some claims lack direct support	Hypothetical or borrowed from literature, not own data	No evidence of failure analysis conducted
Harm pathway	Affected group, mechanism, and severity all named	Population connection present but incomplete for one mode	Vague; specific groups not identified	No harm pathway articulated
Method connection	Explicitly tied to SHAP, LIME, PSI, or security analysis	Methods referenced but linkage not fully shown	Methods mentioned by name only, no connection to findings	No course methods referenced

Before You Present — Self-Check

- I can name at least two distinct failure modes with evidence from my own analysis (not hypothetical).
- For each failure mode I can identify the affected group and explain the mechanism of failure.
- I have connected at least one failure mode to a course method (SHAP, LIME, PSI, or security analysis).
- My failure mode analysis is grounded in the HMDA dataset — not a generic ML discussion.

Q3 — Subgroup Error Measurement (10 points)

Judges want to know: did you measure disparate outcomes across protected groups, and what do the numbers mean?

Criterion	Exemplary 8–10 pts	Proficient 6–7 pts	Developing 3–5 pts	Inadequate 0–2 pts
Metrics computed	At least one of AIR, ME, SMD reported with correct formula	At least one metric computed and correctly applied	Metric attempted but misapplied or reference group unclear	No subgroup metric computed
Groups examined	Multiple protected attributes (race, sex, ethnicity)	At least one protected attribute examined	One group examined, others not acknowledged	No protected attributes examined

Interpretation	Direction, magnitude, and significance correctly explained	Mostly correct; stops short of a full regulatory interpretation	Results presented without meaningful interpretation	No interpretation of any kind
Regulatory framing	Four-fifths rule stated; AIR < 0.80 explicitly flagged and addressed	Four-fifths threshold referenced	Fair-lending significance not explained	No awareness of fair-lending regulatory context

Before You Present — Self-Check

- I have computed at least one of AIR, ME, or SMD across race, sex, or ethnicity.
- I can explain what my bias metric means in plain language to a non-technical audience.
- If any AIR is below 0.80, I flag it explicitly and address it.
- I can explain what the four-fifths rule is and why it matters in mortgage lending.
- I have considered or can speak to intersectional disparities (e.g., race × sex).

Q4 — Residual Risks After Mitigation (3 points)

Judges want to know: what did you try to fix, did it work, and what are you knowingly accepting?

Criterion	Exemplary 3 pts	Proficient 2 pts	Developing 1 pt	Inadequate 0 pts
Mitigation described	At least one mitigation clearly described with rationale	Mitigation described but rationale thin	Mitigation claimed but not substantiated	No mitigation described, or model claimed to be risk-free
Before/after evidence	Quantitative before-and-after comparison shown	Comparison present but incomplete	No before/after evidence provided	No evidence of any kind
Residual risk named	Specific residual risk named — not dismissed	Residual risk acknowledged in general terms	Risk acknowledged only when prompted	Residual risks not disclosed
Conditions stated	Monitoring plan or acceptance conditions articulated	Monitoring referenced but underdeveloped	No conditions or monitoring plan stated	No acknowledgment of ongoing risk

Before You Present — Self-Check

- I can describe at least one mitigation I attempted (e.g., reweighting, threshold adjustment, feature removal).
- I have before-and-after evidence showing what the mitigation achieved.
- I can name at least one residual risk that remains after mitigation.
- I can state the conditions under which the residual risk is acceptable.

Q5 — Deployment Defensibility (2 points)

Judges want to know: should this model be deployed, under what conditions, and what would make you pull it?

Criterion	Exemplary 2 pts	Proficient 1 pt	Developing 0–1 pt	Inadequate 0 pts
Recommendation	Clear, explicit: deploy / do not deploy / deploy with conditions	Recommendation stated but hedged or vague	Recommendation absent or contradicts earlier findings	No deployment recommendation given

Consistency with evidence	Fully supported by Q1–Q4 findings	Mostly consistent; minor gaps between evidence and conclusion	Evidence and conclusion inconsistent	Presentation ends without synthesis
GitHub as audit record	Referenced as a governance artifact	Mentioned but role not articulated	GitHub mentioned only as a submission requirement	GitHub not mentioned
Re-evaluation trigger	At least one specific trigger condition named	Trigger implied but not explicitly stated	No re-evaluation trigger discussed	No monitoring or governance discussed

Before You Present — Self-Check

- My presentation ends with a clear, explicit deployment recommendation (deploy / do not deploy / deploy with conditions).
- My recommendation is consistent with the evidence I presented in Q1–Q4.
- I reference the GitHub repo as part of the audit record — not just a code submission.
- I can state at least one condition that would trigger a model re-evaluation or shutdown.

Presentation Tips: Talking to Your Judges

Your three judges bring professional expertise in nonprofit leadership, civil rights law, and financial technology. None of them will be evaluating the correctness of your Python code. They will be evaluating the quality of your reasoning, the honesty of your analysis, and your ability to communicate technical findings to a non-technical audience — exactly the skills a responsible ML practitioner needs in practice.

Who Your Judges Are

President & CEO, Nonprofit Organization

This judge is likely focused on community impact and equity. They will respond to concrete examples of who is helped or harmed by your model. Ground your bias metrics in human terms: 'An AIR of 0.72 means that for every 10 white applicants approved, only 7 Black applicants with similar profiles are approved.' Avoid jargon without translation.

Senior Counsel, Civil Rights Organization

This judge will be attentive to protected class analysis, the four-fifths rule, and whether your team understands the legal stakes of disparate impact in mortgage lending. Be precise about which groups you analyzed, which metrics you used, and what a regulator would see in your numbers. Do not understate findings.

Founder & CEO, Fintech Company

This judge understands ML systems in production. They will probe your failure modes and deployment defensibility. Be specific about what breaks, when, and why. Have a clear answer to: 'If I deployed this model tomorrow and something went wrong, what is most likely to go wrong and how would you know?' Reference your GitHub repo as operational evidence.

Fifteen Minutes — Suggested Time Allocation

Section	What to Cover	Time
Opening	Model overview, dataset, label definition, team roles	1–2 min
Q1 — Objective	Metric, threshold, and rationale	2 min
Q2 — Failure Modes	Two evidenced failure modes with harm pathways	3 min
Q3 — Subgroup Errors	Bias metrics, groups, regulatory interpretation	3 min
Q4 — Residual Risks	Mitigation attempted, before/after, what remains	2 min
Q5 — Defensibility	Deployment recommendation and conditions	2 min
Close	GitHub repo reference, invitation for questions	1 min

Common Mistakes to Avoid

- Presenting accuracy alone as evidence of model quality. Judges who know fair-lending law will immediately ask about subgroup performance.
- Claiming the model is 'unbiased' or 'fair' without showing the numbers. Honesty about measured disparity scores higher than an unsupported claim of fairness.
- Describing failure modes in generic terms ('the model might not generalize'). Judges want specific, evidenced failures from your HMDA analysis.
- Ending without a deployment recommendation. Every presentation must conclude with a clear answer to: should this model be deployed?
- Treating the GitHub repo as a submission checkbox. Reference it as the audit record — the reproducible evidence behind your claims.
- Using acronyms without translation (AIR, SMD, SHAP, PSI). Define each term the first time you use it.

Q&A: Anticipate These Questions

Likely Judge Question	What a Strong Answer Includes
Why did you choose that metric and not accuracy?	Name the limitation of accuracy on imbalanced data; connect to which error type is costlier for applicants.
What happens to a Black applicant who is wrongly denied?	Describe the harm concretely: delayed homeownership, credit impact, and link to your AIR finding.
If your AIR is below 0.80, why are you recommending deployment?	Explain the mitigation attempted, what it achieved, and the monitoring conditions under which deployment proceeds.
How would you know if the model starts performing worse post-deployment?	Name a specific drift metric (PSI, KS) and a threshold that would trigger a review.
Can I see the code behind this analysis?	Point to your public GitHub repo and describe what a reviewer would find there.

Remember: a good audit is not a perfect model. It is a justified model with documented risks.